

INYOUNG • CLASS 2

Explaining the Paper, Start to Finish

Cao et al., Nature 2025 — Integration of hunger and hormonal state gates infant-directed aggression

HOW TODAY WORKS

Four Boxes, Every Figure

1 ASKED

The question here is whether ____.

2 DID

To test this, they ____.

3 FOUND

What they found was ____.

4 SO WHAT

This tells us ____.

TENSE

What they did, what they found → **past**

They took food away for six hours.

What it means → **present**

This tells us that hunger changes...

VOICE • out loud, active is easier to follow

✓ **They knocked out the receptor in the MPOA.** △ The receptor was knocked out in the MPOA. *fine in a paper, harder to follow in a room*

Figure + question

Each piece starts from one real panel. Point at it while you talk.

Say it to undergrads

Paper term on top, classroom words below. Use the bottom line.

Say it right

Stress the gold syllable. We say these out loud first.

Your line

Your best sentence, corrected, goes here. It becomes your L&S.

Build it up

After each figure, start again from the top.

Last round

The whole paper, once, no stopping.

Why Ask This at All?

Why would a hungry mouse attack a pup it would normally look after?

SET IT UP IN THREE LINES

1 Normally, ____.

2 But animals feel many things at once — ____.

3 So the big question here is ____.

1 WHO

Virgin female mice — ____.

2 WHAT THEY USUALLY DO

They usually ____ or ____ pups.

3 WHAT CHANGES

When they're hungry, ____.

4 WHY IT MATTERS

It shows how two states ____.

SAY IT RIGHT **VIR**gin · be**HA**viour · **PUP** · or**THO**gonal

YOUR LINE · best sentence, corrected

SAY IT TO UNDERGRADS paper term → classroom words

internal state

→ what's going on inside the body — hunger, hormones

social behaviour

→ how an animal treats another animal

virgin female mice

→ female mice that have never had pups

pup-directed care

→ looking after pups

pup-directed aggression

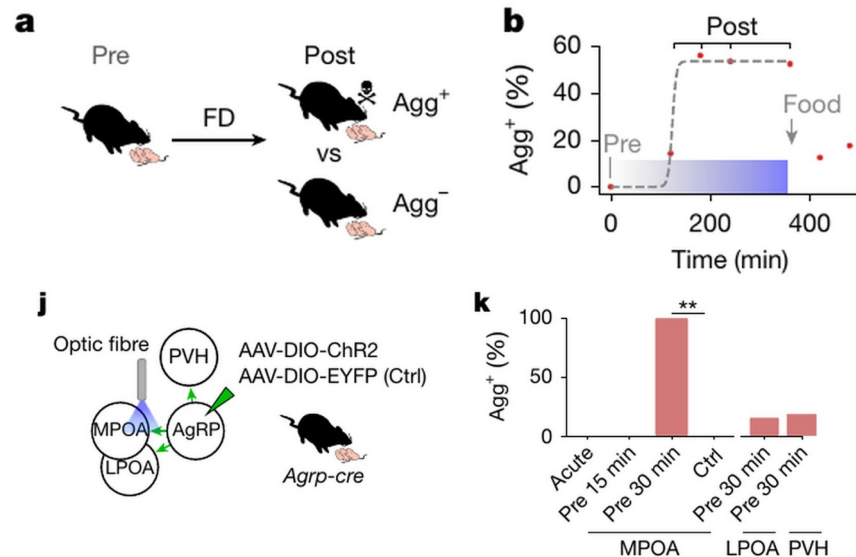
→ attacking pups

orthogonal states

→ two states that normally have nothing to do with each other

Hunger Flips the Switch

How do you prove hunger causes the attack — not stress, not something else?



Cao et al., Nature 648, 138 (2025) • CC BY 4.0 • Fig. 1a,b,k

1 ASKED

The question here is whether ____.

2 DID

To test this, they ____.

3 FOUND

What they found was ____.

4 SO WHAT

This tells us ____.

SAY IT RIGHT hypo**THA**lamus • ag**GRESS**ion • depri**VA**tion • opto**GE**netic

YOUR LINE • best sentence, corrected

SAY IT TO UNDERGRADS paper term → classroom words

food deprivation (FD)

→ taking their food away for six hours

Agg⁺ / Agg⁻

→ the mice that attacked / the ones that didn't

ArcAgRP neurons

→ the brain's hunger neurons

chemogenetic activation

→ switching those neurons on with a drug

optogenetic stimulation

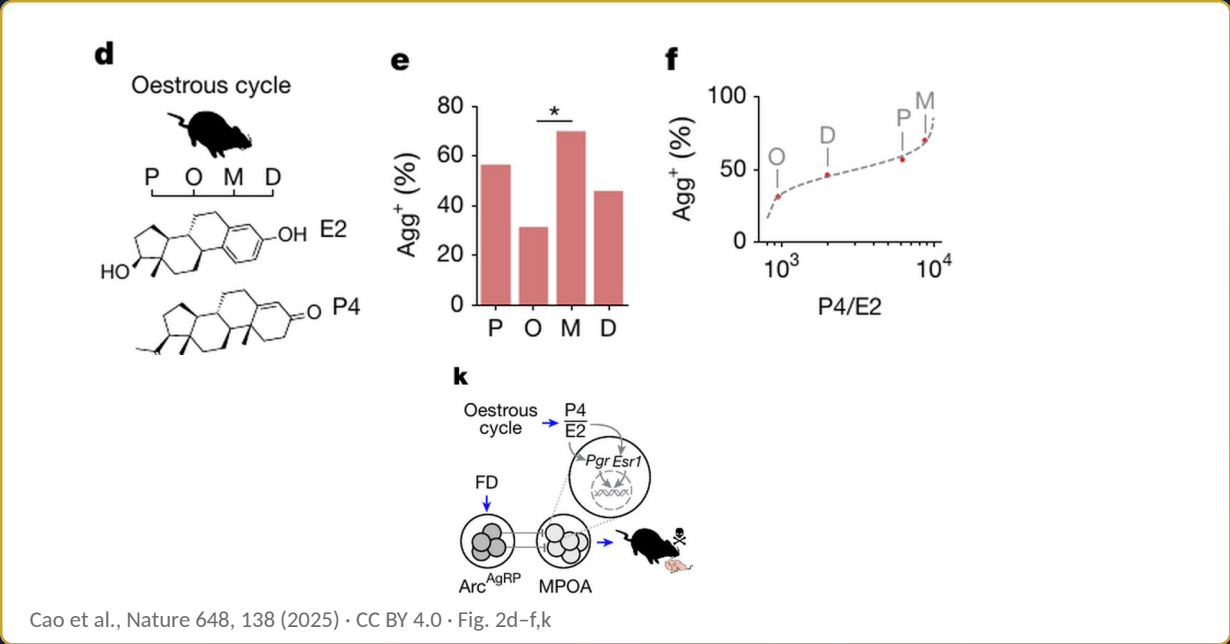
→ switching one pathway on with light

MPOA

→ the part of the hypothalamus that runs parenting

Hormones Set the Odds

If hunger causes it, why did only about 60% of the mice attack?



1 ASKED

The question here is whether ____.

2 DID

To test this, they ____.

3 FOUND

What they found was ____.

4 SO WHAT

This tells us ____.

SAY IT RIGHT OEstrous • oesTRAdiol • proGESTerone • RAtio

YOUR LINE • best sentence, corrected

SAY IT TO UNDERGRADS paper term → classroom words

oestrous cycle

→ the mouse version of the menstrual cycle — four to five days

oestradiol (E2) • progesterone (P4)

→ the two main female sex hormones

P4/E2 ratio

→ not how much of each — the balance between them

receptor knockout

→ removing what the hormone binds to, in one brain area

switching rate

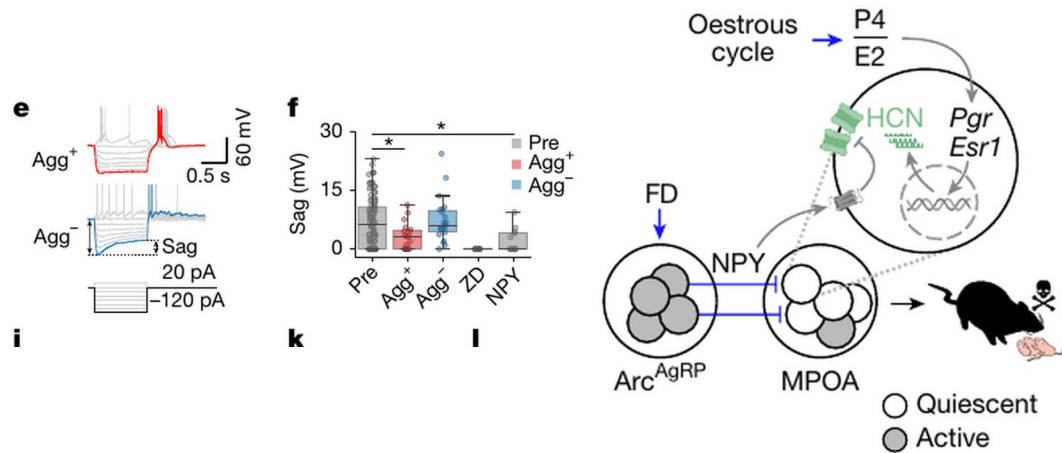
→ the share of mice that flip to attacking

metestrus • oestrus

→ the stages with the highest (70%) and lowest (32%) rate

Where the Two Signals Meet

Two different signals, one behaviour. Where do they actually meet?



Cao et al., Nature 648, 138 (2025) • CC BY 4.0 • Fig. 3e,f,w

1 ASKED

The question here is whether ____.

2 DID

To test this, they ____.

3 FOUND

What they found was ____.

4 SO WHAT

This tells us ____.

SAY IT RIGHT excitability • NEURO peptide Y • CHANNEL • VOLTage

YOUR LINE • best sentence, corrected

SAY IT TO UNDERGRADS paper term → classroom words

patch-clamp recording

→ listening to one neuron's electrical activity

excitability

→ how easily a neuron fires

HCN channels

→ channels that keep a neuron ready to fire

voltage sag

→ the electrical sign that HCN channels are working

NPY (neuropeptide Y)

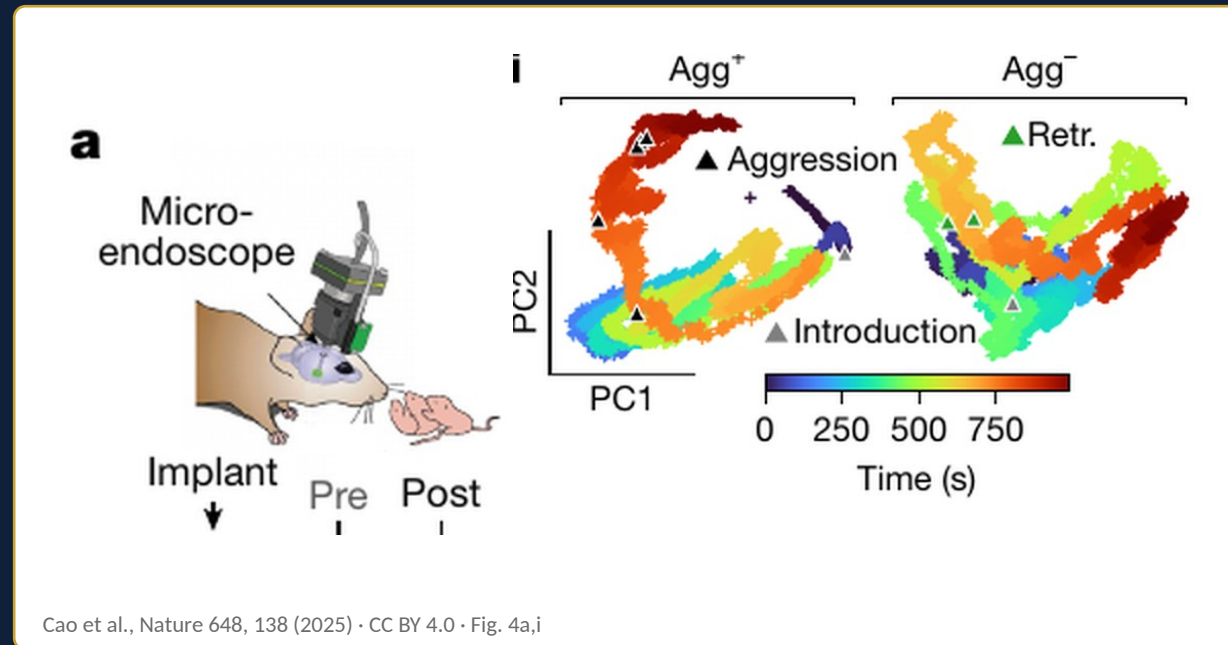
→ the chemical the hunger neurons release

knockdown

→ turning a gene down, not all the way off

Watching It in a Living Brain

All of that was in brain slices. Does it hold in a living, behaving mouse?



1 **ASKED**

The question here is whether ____.

2 **DID**

To test this, they ____.

3 **FOUND**

What they found was ____.

4 **SO WHAT**

This tells us ____.

SAY IT RIGHT Microscope • en**DOS**cropy • **PRIN**cipal • **CAL**cium

YOUR LINE • best sentence, corrected

SAY IT TO UNDERGRADS paper term → classroom words

calcium imaging

→ watching neurons light up when they fire

miniature microscope

→ a tiny microscope mounted on the mouse's head

baseline activity

→ how active the neurons are when nothing is happening

neural state

→ a pattern across many neurons that lasts, not a one-off response

PC1 • PC2

→ a way to squash hundreds of neurons into two lines

decode

→ read the behaviour off the brain activity

The Whole Story

Start from the top. No stopping. Point at each figure as you go.

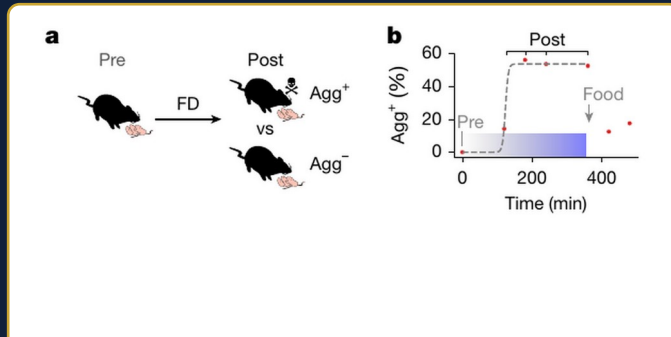


FIG 1 Hunger flips the switch — ____.

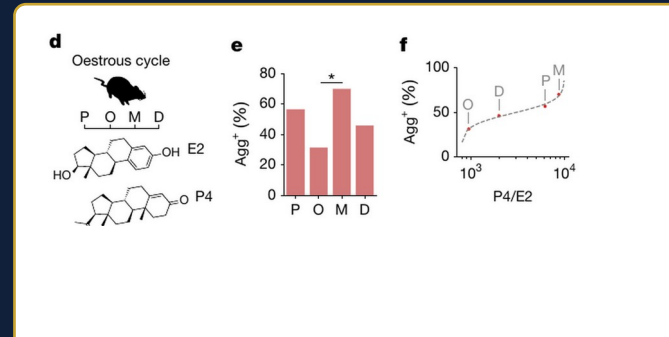


FIG 2 Hormones set the odds — ____.

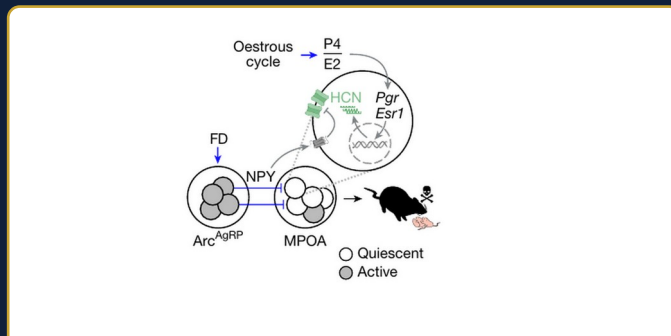


FIG 3 They meet at HCN channels — ____.

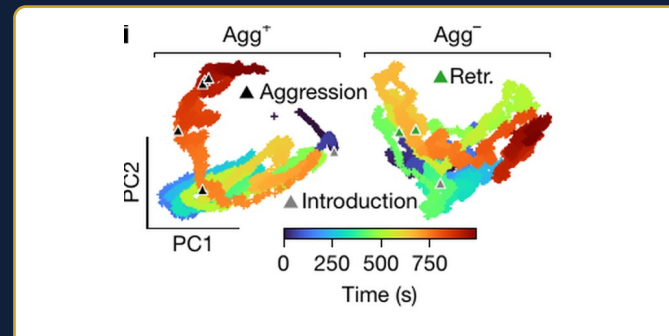


FIG 4 A living brain shows it — ____.

CLOSE IT • pick one

Mechanism

So the same neurons read hunger and hormones, and the behaviour flips when both line up.

Big picture

This is one of the first cases where we can see, cell by cell, how two internal states combine to change a social decision.

For undergrads

What an animal does to another animal depends not just on what it sees, but on what's going on inside it — and now we know one place in the brain where that calculation happens.

TAKE IT TO YOUR LAB

Same Four Boxes, Your Own Research

One minute per box. This is your next lab meeting.

1 ASKED

In my project, I'm asking whether ____.

2 DID

To test this, I ____ / I'm going to ____.

3 FOUND

So far, I've found that ____.

4 SO WHAT

If it holds, it would mean ____.

TENSE, AGAIN • your own work

Done: I measured... **Still going:** I've been running... **Next:** I'm going to... **Meaning:** This suggests...

YOUR LINE • best sentence, corrected

Before Next Class

1

Record it

The three-minute version — about 30 seconds per figure. Send it to Brian.

2

Find out

How long the talk is · the date · whether you make the slides or use your professor's.

3

Listen

L&S 02 is built from today's YOUR LINE sentences — your own words, corrected.